

Daily Progress Report

Week 3_Day3: DOM Manipulation & Event Delegation

Program	3-Month MERN Stack Internship Program
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Week	Week 3 (Day 3)
Date	8 July, 2026

Day 2: DOM Manipulation & Event Delegation

1.1 Topics Studied

- DOM manipulation using JavaScript
- Creating and appending dynamic HTML elements
- Event handling with `addEventListener()`
- Event delegation and event bubbling
- Using `classList.toggle()` to update UI dynamically

1.2 Task Description

Developed a dynamic task manager application where users can add tasks and mark them as completed by clicking on them. The project uses event delegation by attaching a single event listener to the parent `<ul>` element to handle click events on dynamically created `<li>` elements. A delete option was also implemented to remove tasks while using the same event listener.

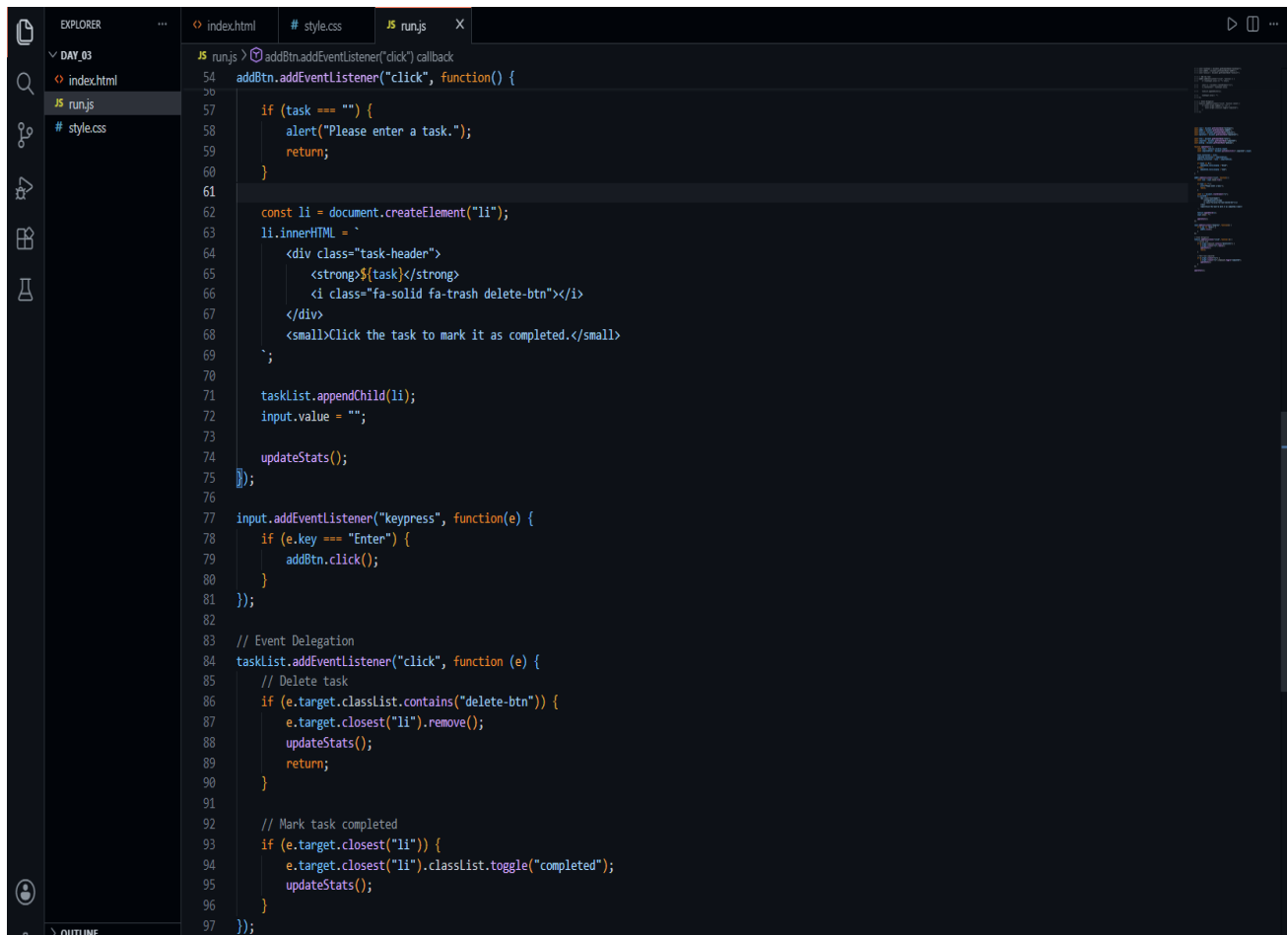
1.3 Implementation Details

- Designed a responsive and modern task manager interface using HTML and CSS.
- Created tasks dynamically using JavaScript and appended them to the task list.
- Implemented event delegation by attaching a single click event listener to the parent `<ul>` element.
- Used `classList.toggle()` to add or remove the completed CSS class when a task is clicked.
- Added a delete button for each task and handled task deletion using the same event listener.
- Displayed live task statistics, including total, completed, and pending tasks.
- Included an empty state message that appears when no tasks are available.

1.4 Outcome

Successfully developed a dynamic task manager application using DOM manipulation and event delegation. The project demonstrated how a single event listener can efficiently manage interactions for dynamically created elements while keeping the code simple and organized. The final application is responsive, interactive, and provides a clean user experience.

Screenshot:



```
54 addBtn.addEventListener("click", function() {
55
56
57     if (task === "") {
58         alert("Please enter a task.");
59         return;
60     }
61
62     const li = document.createElement("li");
63     li.innerHTML = `
64         <div class="task-header">
65             <strong>${task}</strong>
66             <i class="fa-solid fa-trash delete-btn"></i>
67         </div>
68         <small>Click the task to mark it as completed.</small>
69     `;
70
71     taskList.appendChild(li);
72     input.value = "";
73
74     updateStats();
75 });
76
77 input.addEventListener("keypress", function(e) {
78     if (e.key === "Enter") {
79         addBtn.click();
80     }
81 });
82
83 // Event Delegation
84 taskList.addEventListener("click", function (e) {
85     // Delete task
86     if (e.target.classList.contains("delete-btn")) {
87         e.target.closest("li").remove();
88         updateStats();
89         return;
90     }
91
92     // Mark task completed
93     if (e.target.closest("li")) {
94         e.target.closest("li").classList.toggle("completed");
95         updateStats();
96     }
97 });
```

Screenshot:

